

1. Duration + A1c banner

WHAT YOU SEE

A header banner reading DURATION + A1c with an hourglass and a red blood drop — time plus chronic sugar, the two levers driving diabetic kidney scarring.

WHAT IT ENCODES

Diabetic kidney disease (DKD) risk is driven by two modifiable levers: cumulative disease DURATION and chronic hyperglycemia (A1c). DKD develops in ~30-40% of diabetics and is the leading cause of ESRD in the US. Tight glycemic control (A1c target individualized ~7%) plus BP control are the foundations that slow progression. Mnemonic: 'Time + Sugar = Scar.'

BOARD TRAP

Tight glucose control reduces MICROvascular complications (nephropathy, retinopathy, neuropathy) far more reliably than macrovascular events — the board distractor claims intensive A1c lowering primarily prevents MI/stroke, which is not well supported (ACCORD even showed harm with very aggressive targets).

2. Hyperfiltration / GFR 150

WHAT YOU SEE

GFR gauge pinned at 150 labeled HYPERFILTRATION

WHAT IT ENCODES

The EARLIEST renal change in diabetes is glomerular HYPERFILTRATION — GFR rises to ~150 mL/min before any albumin appears. Mechanism: hyperglycemia causes afferent arteriolar vasodilation and efferent vasoconstriction (angiotensin II-mediated), raising intraglomerular pressure. This is exactly why ACEi/ARB (efferent dilation) and SGLT2i (restore tubuloglomerular feedback at the macula densa) are protective. Mnemonic: high pressure today = scarring tomorrow.

BOARD TRAP

Early DKD shows a SUPRANORMAL (high) GFR, not a low one — the classic wrong answer is to pick 'declining GFR' as the first detectable change. GFR only falls later, after albuminuria is established.

3. Hypertrophy

WHAT YOU SEE

Enlarged kidney labeled HYPERTROPHY

WHAT IT ENCODES

In the first 0-5 years the kidneys physically ENLARGE (hypertrophy) and hyperfilter — a reversible adaptive response to glucose-driven hemodynamic stress. On imaging the diabetic kidney is often normal-to-large even in advanced CKD, unlike most chronic kidney diseases where kidneys shrink.

BOARD TRAP

Small, shrunken kidneys on ultrasound argue AGAINST diabetic nephropathy and point to a different chronic process — in DKD the kidneys characteristically stay normal-sized or enlarged despite falling GFR.

4. Pericyte loss / GBM thickening

WHAT YOU SEE

Stone arch labeled GBM

WHAT IT ENCODES

The structural hallmarks are glomerular basement membrane (GBM) THICKENING and mesangial matrix expansion, culminating in nodular glomerulosclerosis — the KIMMELSTIEL-WILSON nodules (PAS-positive acellular nodules), the pathognomonic biopsy finding of diabetic nephropathy. Efferent arteriolar hyalinosis is also characteristic.

BOARD TRAP

Kimmelstiel-Wilson nodules are specific for diabetic nephropathy — do NOT confuse them with the 'spike-and-dome'/subepithelial deposits of membranous nephropathy or the wire-loops of lupus nephritis.

5. Micro-albuminuria (UACR >30)

WHAT YOU SEE

A few coins trickling through the wall at a sign reading MICRO, UACR more than 30 mg per g — the first small albumin leak through the filtration barrier.

WHAT IT ENCODES

Moderately increased albuminuria — urine albumin-to-creatinine ratio (UACR) 30-300 mg/g — is the EARLIEST clinical/lab marker of DKD and an independent cardiovascular risk marker. Screen with a spot UACR yearly: start at diagnosis in T2DM and at 5 years in T1DM. Confirm 2 of 3 abnormal samples over 3-6 months (exercise, fever, infection, menses can falsely elevate it).

BOARD TRAP

A standard urine DIPSTICK does not detect microalbuminuria — it only turns positive at macroalbuminuria (>300 mg/g). The board trap is using a negative dipstick to 'rule out' early DKD; you must order a quantitative UACR.

6. Macro-albuminuria

WHAT YOU SEE

Gold coins spilling through the breached GBM wall into the river labeled MACRO — heavy albumin leak (UACR >300) pouring out as overt proteinuria.

WHAT IT ENCODES

Severely increased albuminuria (UACR >300 mg/g, i.e. overt proteinuria) marks overt nephropathy and predicts progressive GFR decline toward ESRD. It can reach nephrotic-range and is a strong driver of cardiovascular mortality. Lowering albuminuria with ACEi/ARB + SGLT2i is itself a treatment goal.

BOARD TRAP

Macroalbuminuria reflects chronic glomerular injury and does not regress to normal simply with a few weeks of good sugar control — the trap is expecting albuminuria to disappear with glycemic control alone; RAAS blockade is required to reduce it.

7. Timeline year markers

WHAT YOU SEE

Signposts YEAR 0–5, 5–10, 15–25 with falling GFR 60 then <10

WHAT IT ENCODES

Natural history of T1DM nephropathy: hyperfiltration/hypertrophy (years 0-5) → microalbuminuria (5-10) → macroalbuminuria with falling GFR (10-20) → ESRD (~15-25). GFR <60 defines CKD stage 3; GFR <15 (here <10) is stage 5/ESRD requiring renal replacement. T2DM may already be at any stage at diagnosis because onset is silent.

BOARD TRAP

Do not assume a newly diagnosed T2DM patient is at 'year 0' of kidney disease — because T2DM smolders undiagnosed for years, screen for albuminuria and reduced GFR AT THE TIME OF DIAGNOSIS, not 5 years later as in T1DM.

8. Albumin soldier + ACEi/ARB shield

WHAT YOU SEE

Knight holding ALBUMIN with ACEi ARB shield and down arrow

WHAT IT ENCODES

ACE inhibitors (e.g. lisinopril 10-40 mg/day) or ARBs (e.g. losartan 50-100 mg, valsartan 80-320 mg) are FIRST-LINE in DKD with albuminuria and/or hypertension. They dilate the efferent arteriole, lower intraglomerular pressure, reduce albuminuria, and slow progression independent of BP lowering. Titrate to the maximally tolerated dose.

BOARD TRAP

Do NOT combine an ACEi WITH an ARB — dual RAAS blockade increases hyperkalemia, AKI, and hypotension without benefit (ONTARGET/VA NEPHRON-D). Likewise, do not add a direct renin inhibitor (aliskiren) to an ACEi/ARB in diabetics (ALTITUDE harm). Use ONE agent.

9. K⁺ / Cr + angioedema cat

WHAT YOU SEE

Cat labeled K⁺ Cr with ANGIOEDEMA

WHAT IT ENCODES

Key ACEi/ARB adverse effects: HYPERKALEMIA, a transient creatinine RISE, and (with ACEi) cough and ANGIOEDEMA (bradykinin-mediated). Check a basic metabolic panel 1-2 weeks after starting or up-titrating. Manage hyperkalemia with diet, diuretics, or potassium binders (patiomer) rather than stopping a beneficial drug when possible.

BOARD TRAP

A creatinine rise of UP TO ~30% from baseline after starting an ACEi/ARB is expected and acceptable — do NOT discontinue the drug for this. Only stop/investigate if creatinine rises >30%, suggesting bilateral renal artery stenosis or volume depletion. (Angioedema, however, mandates immediate permanent discontinuation.)

10. T2DM CKD↓ guard + SGLT2

WHAT YOU SEE

Green guard labeled T2DM CKD↓ pointing to SGLT-2 sign

WHAT IT ENCODES

SGLT2 inhibitors (empagliflozin 10 mg, dapagliflozin 10 mg, canagliflozin 100 mg) slow CKD progression, reduce albuminuria, and cut cardiorenal events/hospitalization in T2DM — added ON TOP of a maximally dosed ACEi/ARB. They restore tubuloglomerular feedback, lowering intraglomerular pressure. Expect a small initial 'dip' in eGFR that is hemodynamic and reverses; continue the drug.

BOARD TRAP

SGLT2 inhibitors are now used for renal protection down to relatively low eGFR (~20-25), and dapagliflozin/empagliflozin benefit even non-diabetic CKD — the trap is stopping them just because eGFR is modestly reduced or because the patient is at goal A1c. Watch for euglycemic DKA and genital mycotic infections.

11. Treatment tower (GLP-1 / SGLT2 / ACEi-ARB / BP)

WHAT YOU SEE

Tower signs: GLP-1 IF T2DM + CVD, SGLT-2, ACEi or ARB IF ALBUMINURIA, BP <130/80

WHAT IT ENCODES

DKD management stack: (1) ACEi OR ARB if albuminuria/HTN; (2) SGLT2 inhibitor for cardiorenal protection; (3) GLP-1 RA (semaglutide, dulaglutide, liraglutide) preferred when ASCVD/obesity present and as add-on glucose control; (4) BP target <130/80; (5) consider finerenone (nonsteroidal MRA) in T2DM with albuminuric CKD to further lower CV/renal events on top of ACEi/ARB+SGLT2i. Statin therapy and glycemic control round it out.

BOARD TRAP

Don't pick a sulfonylurea or basal insulin as the 'renoprotective' add-on — they lower glucose but lack the proven cardiorenal benefit of SGLT2i/GLP-1 RA. Among agents, SGLT2i and GLP-1 RA are the disease-modifying choices in DKD.

12. Think GN / vascular detective

WHAT YOU SEE

Detective labeled think GN or VASCULAR

WHAT IT ENCODES

Diabetic nephropathy is usually a CLINICAL diagnosis (longstanding diabetes + albuminuria + retinopathy, no biopsy needed). But play detective: if the picture is atypical, consider glomerulonephritis, vascular disease, or another superimposed renal disease and pursue workup/biopsy.

BOARD TRAP

An active urine sediment (dysmorphic RBCs, RBC casts) is NOT a feature of diabetic nephropathy and signals a glomerulonephritis instead — the trap is attributing hematuria with casts to diabetes; that demands a separate glomerular workup.

13. NO retinopathy / atypical signs

WHAT YOU SEE

Red sign NO RETINOPATHY, CKD, OTHER DX with nephro referral list (eGFR, HEMATURIA, NEPHROTIC)

WHAT IT ENCODES

Features that should make you suspect NON-diabetic kidney disease and refer to nephrology: absence of diabetic RETINOPATHY, rapidly falling eGFR, active urinary sediment/hematuria, abrupt nephrotic-range proteinuria, short diabetes duration, or refractory hypertension. Retinopathy and nephropathy are tightly linked microvascular siblings.

BOARD TRAP

In a T1DM patient, the ABSENCE of retinopathy argues strongly AGAINST diabetic nephropathy (the two co-travel) — the board trap is calling proteinuria 'diabetic' in a T1 patient with clear retinas; you must investigate an alternative cause. (The association is somewhat weaker in T2DM.)

14. Protein restriction sign

WHAT YOU SEE

Sign 0.8 g per kg per day labeled PROTEIN

WHAT IT ENCODES

Moderate dietary protein restriction to ~0.8 g/kg/day is recommended in non-dialysis DKD to reduce glomerular hyperfiltration and slow decline. Pair with sodium restriction (<2 g/day) to potentiate RAAS blockade and BP control.

BOARD TRAP

Do NOT severely restrict protein (e.g. <0.6 g/kg/day) — over-restriction causes malnutrition and muscle wasting without added renal benefit, and protein needs INCREASE once a patient is on dialysis (~1.2 g/kg/day). The 0.8 g/kg/day figure applies to pre-dialysis CKD.